

REVIEW OF PHARMACOTHERAPY OPTIONS FOR THE TREATMENT OF ATTENTION-DEFICIT/HYPERACTIVITY DISORDER (ADHD) AND ADHD-LIKE SYMPTOMS IN CHILDREN AND ADOLESCENTS WITH DEVELOPMENTAL DISORDERS

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Developmental disorders such as subaverage intelligence, pervasive developmental disorders, and genetic syndromes are frequently associated with comorbid attention-deficit/hyperactivity disorder (ADHD) or ADHD-like symptoms. While there are not pharmacological cures for these developmental disorders, coinciding ADHD and ADHD-like symptoms that contribute to difficulties in psychosocial functioning are frequently able to be addressed by pharmacotherapy. This article reviews what is known about the efficacy and tolerability of pharmacological interventions for the treatment of children and adolescents suffering from developmental disorders and comorbid ADHD/ADHD-like symptoms.

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Key Words: developmental disorder; subaverage intelligence; attention-deficit/hyperactivity disorder; treatment

With United States community sample prevalence estimates ranging from ~2% up to 12%, attention-deficit/hyperactivity disorder (ADHD) is the most common neurodevelopmental disorder in typically-developing children [Kratochvil et al., 2009]. Of concern is the increased rate at which ADHD and ADHD-like symptoms—hyperactivity, impulsivity, and inattention—occur in children with developmental delays. It has been reported that between 6% [Capone et al., 2006; Dykens, 2007] and 80% [Frazier et al., 2001] of children with developmental delays including genetic disorders, subaverage intelligence, pervasive developmental disorders, and complications owing to prenatal exposure to ethanol suffer from comorbid ADHD/ADHD-like symptoms. Pharmacological treatment studies of ADHD and ADHD-like symptoms in these vulnerable populations have primarily focused on stimulants; however, atomoxetine, antidepressants, alpha adrenergic agonists, and antipsychotics have been reported to be beneficial in the reduction of ADHD symptomatology.

In typically-developing children, ADHD is associated with significant familial, social, and academic impairment [Kratochvil et al., 2009]. Pharmacotherapy is frequently a first line treatment for typically-developing children with ADHD,

and there is a strong evidence base to support the use of several medications, particularly stimulant medications [Pliszka et al., 2007; Findling, 2008; Kratochvil et al., 2009]. Stimulants are not only helpful in reducing ADHD symptomatology, but also are usually tolerable and the adverse effects are frequently without serious medical sequelae [Pliszka et al., 2007]. Stimulant treatment research in very young children with ADHD (ages 3–5 years), however, has reported that methylphenidate may be associated with a greater side effect burden than older children with ADHD [Wigal et al., 2006]. Of note, nonstimulant medications such as atomoxetine and extended-release guanfacine have been reported to be useful for the treatment of ADHD in typically developing children and adolescents [Pliszka et al., 2007; Findling, 2008; Conner and Rubin, 2010].

Despite the cooccurrence of ADHD and ADHD-like symptoms within populations of children with developmental delays, subaverage intelligence and pervasive developmental disorder, children with these conditions are frequently excluded from ADHD treatment studies. Additionally, according to DSM-IV-TR criteria, a diagnosis of ADHD may not be given in the presence of a pervasive developmental disorder. Interestingly, this limitation is likely to change with the publication of DSM-V. Further, the data for the treatment of ADHD and ADHD-like symptoms within these populations vary. Thus, a review of ADHD treatments for pediatric patients suffering from developmental disorders is warranted.

A PubMed search was conducted for topical literature published between 1990 and 2010. Search terms included, but were not limited to: attention-deficit/hyperactivity disorder, developmental disorder, developmental disability, subaverage

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intelligence, and pervasive developmental disorder. Additionally, published review articles on similar topics were consulted for primary sources.

DEVELOPMENTAL DISABILITIES/SUBAVERAGE INTELLIGENCE

Several placebo-controlled studies for the treatment of comorbid ADHD in children with subaverage intelligence quotient (IQ) have been conducted. The subaverage IQs of these children were not noted to be associated with known genetic disorders. While many of these trials have been characterized by a crossover design, they are generally limited by small sample sizes. However, regardless of design and sample size, these studies have reported salutary treatment effects within this population. A summary of selected trials for the treatment of comorbid ADHD and ADHD-like symptoms in children with subaverage intelligence is listed in Table 1.

Methylphenidate

Methylphenidate is perhaps the most extensively studied medication in children and adolescents suffering from ADHD and comorbid subaverage intelligence. In 1990, Handen et al. published the first of four studies of the efficacy of methylphenidate in this population. Twelve children (aged 6–9 years) with IQ scores of 50–74 who met DSM-III-R diagnostic criteria for ADHD participated in a 6-week, double-blind crossover study of the efficacy of two doses of methylphenidate compared to placebo in which each dose was given twice daily for a 1-week period. Both doses of methylphenidate, 0.3 and 0.6 mg kg⁻¹ were associated with a positive response on the Conners Hyperactivity Index in 75% of these patients. Similar results were found by Handen et al. [1992] in a 6-week replication study of 14 children (aged 6–12 years) with IQ scores of 48 to 74 and comorbid ADHD. As in the previous study, each dose of methylphenidate or placebo was administered twice daily for a 1-week period. Sixty-four percent of the patients were considered methylphenidate responders at total daily doses of 0.6 and 1.2 mg kg⁻¹, based upon the Conners Hyperactivity Index.

A third study by Handen et al. [1995] explored the effects of two methylphenidate doses (0.3 and 0.6 mg kg⁻¹) compared to placebo on independent play and academic functioning in 22 children (ages 6–12 years) with subaverage IQ and ADHD. Exclusion

criteria included a diagnosis of autism or pervasive developmental disorder. After two initial weeks of baseline procedures, each dose of study medication was given twice daily for periods of 1 week each. The authors noted that the efficacy of treatment with methylphenidate was measurable by clinic-based assessment tools using both independent play and academic task paradigms. Treatment with 0.6 mg kg⁻¹ of methylphenidate provided the greatest benefits on variables such as decreases in physical movement and improvements in on-task behavior. In a fourth study, Handen et al. [1999] examined the same doses of methylphenidate (0.3 and 0.6 mg kg⁻¹) compared to placebo in 11 preschool-aged children (4–5 years) with developmental disabilities and ADHD or long-term documented concerns with inattention and overactivity. IQ ranged from 48 to 78, and a diagnosis of autism/pervasive developmental disorder was ruled out for all patients. The first week of the study was devoted to completing baseline measures; during the next 3 weeks of the study, each medication dose was administered twice daily for 1-week periods. Treatment with methylphenidate was superior to placebo in eight of the 11 children, as indicated by a minimum 40% decrease in teacher-rated Conners Hyperactivity Index and/or Hyperactive-Distractible subscale of the Behar Preschool Behavior Questionnaire.

While all of these studies concluded that treatment with methylphenidate was beneficial in this population at a rate similar to typically-developing children with ADHD, significant adverse events such as severe social withdrawal, increased crying, drowsiness, and irritability were noted, especially at the higher dose (0.6 mg kg⁻¹) of methylphenidate [Handen et al., 1990, 1999]. These significant adverse events are similar to those behavioral adverse events frequently reported in very young children with ADHD during treatment with methylphenidate [Wigal et al., 2006].

In 1991, Aman et al. evaluated the efficacy of methylphenidate and thioridazine in 30 children (aged 4–16 years) with subaverage IQs and psychiatric diagnoses of DSM-III attention deficit disorder (80%) or conduct disorder (10%). One child was diagnosed with a pervasive developmental disorder in addition to symptoms of attention deficit disorder with hyperactivity. These patients received 3 weeks each of treatment with placebo, methylpheni-

date (0.40 mg kg⁻¹ day⁻¹), and thioridazine (1.75 mg kg⁻¹ day⁻¹). The phenothiazine thioridazine was probably selected owing to two factors. First, thioridazine has an historic place as one of the first drugs to treat hyperactivity in typically developing children. In addition, it was commonly used to treat children and adults with subaverage intelligence. Compared to both thioridazine and placebo, treatment with methylphenidate was associated with greater response. A similar double-blind crossover study was conducted by Aman et al. [1993] in which 28 children (aged 5–13 years) with subaverage intelligence (IQ ≤ 75) and comorbid ADHD received four weeks each of placebo, methylphenidate (0.4 mg kg⁻¹ day⁻¹), and fenfluramine (gradually increased to 1.5 mg kg⁻¹ day⁻¹). The serotonin depleting agent fenfluramine had shown promise for possibly reducing hyperactivity, inattention, and stereotypic behavior in pediatric patients with autism. Fenfluramine likely was chosen as an active treatment owing to the behavioral symptom similarities between pediatric patients with autism and this particular population. Both active treatments were beneficial for the improvement of ADHD symptoms, but in different ways. Specifically, methylphenidate alone was associated with improvements on the Inattention subscale of the Conners Teacher Rating Scale, while fenfluramine alone was associated with improvements on the Irritability and Inappropriate Speech subscales of the parent-rated Aberrant Behavior Checklist as well as the Conners Abbreviated Symptom Questionnaire. Both methylphenidate and fenfluramine were associated with improvements on the teacher-rated Conduct Problem, Hyperactivity, and Irritability subscales, as well as the parent-rated subscales of Hyperactivity, Motor Excess, and Conduct problem. Therefore, it was suggested that, at the time, both of these medications could have a useful role in the treatment of ADHD in children with subaverage intelligence.

The effectiveness of three doses of methylphenidate (0.15, 0.3, and 0.6 mg kg⁻¹, administered twice daily) compared to placebo was investigated by Pearson et al. [2003] in a double-blind, placebo-controlled crossover study. Twenty-four children [mean age of 10.9 (2.4) years] with subaverage intelligence [mean IQ 56.5 (10.4)] and ADHD without any additional comorbid psychiatric diagnoses participated in the

Table 1. Summary of Selected Studies for the Treatment of Comorbid ADHD and ADHD-like Symptoms in Children with Subaverage Intelligence

Authors (year)	Sample Size and Age Range or Mean (SD) Age	Study Design	Medication and Total Daily Dose	Primary ADHD Outcome Measures	Improvement/Effectiveness/Response
Handen et al. (1990)	N = 12; 6–9 years	6-week double-blind, placebo-controlled crossover	Methylphenidate 0.6, 1.2 mg kg ⁻¹	Conners Teacher Rating Scale—Hyperactivity Index	9/12 (75%) improved on either dose
Handen et al. (1992)	N = 14; 6–12 years	6-week double-blind, placebo-controlled crossover	Methylphenidate 0.6, 1.2 mg kg ⁻¹	Conners Teacher Rating Scale—Hyperactivity Index	9/14 (64%) considered medication responders
Handen et al. (1999)	N = 11; 4–5 years	4-week double-blind, placebo-controlled crossover	Methylphenidate 0.6, 1.2 mg kg ⁻¹	Conners Teacher Rating Scale—Hyperactivity Index; Preschool Behavior Questionnaire—Hyperactive-Distractible subscale	8/11 (73%) considered medication responders
Aman et al. (1991)	N = 30; 4–16 years	9-week double-blind, placebo-controlled crossover	Thioridazine, 1.75 mg kg ⁻¹ ; Methylphenidate 0.4 mg kg ⁻¹	Conners Teacher Questionnaire; Conners Abbreviated Teacher Rating Scale; Revised Behavior Problem Checklist	11/30 (37%) considered methylphenidate responders; 8/30 (27%) considered thioridazine responders
Aman et al. (1993)	N = 28; 5–13 years	12-week double-blind, placebo-controlled crossover	Methylphenidate 0.4 mg kg ⁻¹ ; Fenfluramine 1.5 mg kg ⁻¹ (max)	Conners Teacher Questionnaire; Aberrant Behavior Checklist; Revised Behavior Problem Checklist; Abbreviated Symptom Questionnaire	15/28 (54%) considered methylphenidate responders (not calculated for fenfluramine)
Pearson et al. (2003)	N = 24; 10.9 (2.4) years	4-week double-blind, placebo-controlled crossover	Methylphenidate 0.3, 0.6, 1.2 mg kg ⁻¹	ADD-H Comprehensive Teacher Rating Scale; Conners Teacher Rating Scale; Conners Parent Rating Scale; Aberrant Behavior Checklist; Revised Behavior Problem Checklist; Personality Inventory for Children-Revised (Hyperactivity and Delinquency subscales)	Teacher ratings reported steady decline of both symptoms of inattention and hyperactivity with ascending methylphenidate doses
Correia-Filho et al. (2005)	N = 45; 6–16 years	4-week single-blind, parallel-group	Risperidone, 4 mg (max); Methylphenidate 0.7 mg kg ⁻¹ (max)	SNAP-IV subscales: Total; Inattentive; Hyperactive/impulsive; Oppositional	Risperidone associated with higher effect sizes for Total, Hyperactivity, and Oppositional scores; methylphenidate associated with higher effect size for Inattentive scores
Agarwal et al. (2001)	N = 10; 7.6 (0.54) years	12-week double-blind, placebo-controlled crossover	Clonidine, 4, 6, 8 µg kg ⁻¹	Parent Symptom Questionnaire; Hillside Behavior Rating Scale	All patients met criteria for improvement (a rating of 1 on the gross motor activity item of the Hillside Behavior Rating Scale and a rating of <i>not at all</i> on the Parent Symptom Questionnaire items “constantly fidgeting,” “always climbing,” “unable to stop a repetitive activity,” and “acts as if driven by a motor”)

ADHD = attention-deficit/hyperactivity disorder; ADD-H = attention deficit disorder with hyperactivity.

trial. The most significant behavioral improvements were noted at the 0.6 mg kg⁻¹ methylphenidate dose for teacher ratings of inattention, hyperactivity, aggression, and asocial behavior on the ADD-H Comprehensive Teacher Rating Scale and the Conners Teacher Rating Scale. Notably, no significant improvements occurred at the 0.15 mg kg⁻¹ dose of methylphenidate compared to placebo.

Correia-Filho et al. [2005] evaluated the short-term efficacy and tolerability of risperidone and methylphenidate for the reduction of ADHD symptoms in children and adolescents with subaverage intelligence. Forty-five patients, ages 6–16 years old, meeting DSM-IV criteria for moderate mental retardation and ADHD participated in this randomized, single-blind, parallel group trial. According to ratings on the efficacy measures (the SNAP-IV Rating Scale and the Hyperactivity subscale of the Nisonger Child Behavior Rating Form), both treatment groups experienced a reduction in ADHD symptoms. However, posthoc analyses suggested a more pronounced treatment effect for risperidone. Owing to the side effects associated with risperidone treatment, namely weight gain, the authors posit that children with subaverage intelligence and ADHD may benefit from a trial with methylphenidate first prior to considering a treatment course with risperidone.

Amphetamine-based preparations

One published case study [Payton et al., 1989] reported that a 7-year-old male suffering from DSM-III attention deficit disorder with hyperactivity (ADHD), microcephaly, and an IQ of 66 experienced reductions in excessive movement and increases in on-task behavior during treatment with dextro-amphetamine (doses of 5 mg twice daily and 10 mg twice daily). It does not appear, however, that placebo-controlled treatment studies evaluating amphetamine-based preparations have been executed in children and adolescents suffering from ADHD comorbid with subaverage intelligence in the absence of a diagnosis of a pervasive developmental disorder and/or known genetic syndrome.

Clonidine

One randomized, double-blind, placebo-controlled trial of clonidine has been published in this population. Using a crossover design, Agarwal et al. [2001] evaluated three fixed doses of oral clonidine (4, 6, and 8 mcg kg⁻¹ day⁻¹) in 10 children with subaverage intelligence diagnosed with ICD-10

hyperkinetic disorder (comparable to DSM-IV ADHD-combined type). Treatment with clonidine was initiated at 4 mcg kg⁻¹ day⁻¹ and increased to 6- to 8-mcg doses at the end of the 2nd and 4th week. Each treatment, placebo and clonidine, lasted for 6 weeks. Parent and clinician ratings showed a considerable dose-related response to clonidine in hyperactivity, impulsivity, and inattention. The most common side effect was drowsiness, which was tolerable in most cases.

Pervasive Developmental Disorders

A considerable amount of pharmacological treatment research of methodological rigor has been conducted in children with pervasive developmental disorders (PDD) to address many of the disorders' dysfunctional coinciding behaviors [Leskovec et al., 2008]. The reported rates of ADHD comorbidity in this population are rather high—between 68% [Yoshida and Uchiyama, 2004] and ~80% [Frazier et al., 2001]—and symptoms of hyperactivity, impulsivity, and inattention are common interfering behaviors in children with PDD [Leskovec et al., 2008]. Psychostimulants, especially methylphenidate, are perhaps the most extensively studied class of medication in this population [Aman et al., 2008]. Other medications that have been reported to be beneficial in the treatment of ADHD/ADHD-like symptoms in children with PDD include alpha-2 adrenergic agonists and noradrenergic reuptake inhibitors [Aman et al., 2008]. A summary of selected studies for the treatment of ADHD-like symptoms in pediatric patients with a pervasive developmental disorder diagnosis is listed in Table 2.

Methylphenidate

Quintana et al. [1995] published the first reported double-blind, placebo-controlled crossover study evaluating the efficacy of methylphenidate in children with autistic disorder. After a 2-week baseline period, 10 children (ages 7–11 years) meeting DSM-III-R criteria for autistic disorder were randomly assigned to be treated with methylphenidate (10 or 20 mg, twice daily) or placebo for two 2-week trials. Treatment with methylphenidate was associated with a modest, albeit statistically significant, improvement in hyperactivity over placebo. Notably, neither dose of methylphenidate was associated with significant side effects.

Another double-blind, placebo-controlled crossover study of methylphe-

nidate in this population reported reductions in hyperactivity following treatment with methylphenidate [Handen et al., 2000]. Thirteen children (ages 5–11 years) with autism and ADHD/ADHD-like symptoms participated in this study of two methylphenidate doses: 0.3 and 0.6 mg kg⁻¹. Each medication dose was administered two to three times daily for 7 consecutive days. Eight of the 13 children were considered to be responders to methylphenidate treatment, based upon a decrease of at least 50% in hyperactivity ratings between placebo and methylphenidate. The authors noted that, especially on the 0.6 mg kg⁻¹ dose, significant side effects such as social withdrawal and irritability occurred in almost 40% of their patients.

A large, multi-site, randomized controlled crossover trial of three methylphenidate doses (0.125, 0.250, and 0.500 mg kg⁻¹ day⁻¹) in children with PDD and hyperactivity was conducted by the Research Units on Pediatric Psychopharmacology (RUPP) Autism Network [RUPP, 2005]. Study participants were 72 children (ages 5–14 years) meeting DSM-IV criteria for autistic disorder, Asperger disorder, or PDD not otherwise specified who were suffering from moderate to severe hyperactivity. After a 1-week test-dose phase to assess methylphenidate tolerability, subjects entered a 4-week randomized-order, placebo-controlled, double-blind crossover phase, followed by an 8-week open-label continuation phase for treatment responders. Of the original 72 study participants, 66 were randomized into the crossover phase, 35 of which were considered to be responders to methylphenidate treatment based upon reductions of hyperactivity symptoms and ratings of “much improved” or “very much improved” on the Clinical Global Impressions-Improvement (CGI-I). Compared to placebo, methylphenidate was more efficacious in improving inattention, distractibility, hyperactivity, and impulsivity. The authors noted that, while approximately half of their study population responded to treatment with methylphenidate, these children responded at a rate less than that seen in the treatment of ADHD in typically developing children. Additionally, adverse events were not only more frequent, but also severe enough to lead to study discontinuation in children with PDD and hyperactivity.

Recently, Ghuman et al. [2009] investigated the short-term efficacy and safety of methylphenidate to treat symptoms of ADHD in 14 young children (ages 3–5 years) with PDD or intellec-

Table 2. Summary of Selected Studies for the Treatment of ADHD-like Symptoms in Children with Pervasive Developmental Disorders

Authors (year)	Sample Size and Age Range or Mean (SD) Age	Study Design	Medication and Total Daily Dose	Primary ADHD Outcome Measures	Improvement/Effectiveness/Response
Quintana et al. (1995)	N = 10; 8.5 (1.3) years	6-week double-blind, placebo-controlled crossover	Methylphenidate; 20, 40 mg	Conners Abbreviated Parent-Teacher Questionnaire; Aberrant Behavior Checklist	MPH associated with a statistically significant improvement over placebo on hyperactivity measures
Handen et al. (2000)	N = 13; 5–11 years	3-week double-blind, placebo-controlled crossover	Methylphenidate; 0.6, 1.2 mg kg ⁻¹	Conners Teacher Rating Scale—Hyperactivity Index	8/13 (62%) considered medication responders
RUPP (2005)	N = 72; 5–14 years	1-week test dose, 4-week double-blind, placebo-controlled crossover, 8-week open label continuation	Methylphenidate; 7.5–50.0 mg	Aberrant Behavior Checklist—Hyperactivity subscale	35/72 (49%) considered medication responders
Ghuman et al. (2009)	N = 14; 4.8 (1.0) years	Single-blind titration, 4-week double-blind, placebo-controlled crossover	Methylphenidate; 20 mg (max)	Conners Parent Rating Scale—Revised—DSM-IV-ADHD Subscale; Nisonger Child Behavior Rating Form—Parent-Hyperactive Subscale; Clinical Global Impressions Severity and Improvement; Children's Global Assessment Scale	7/14 (50%) rated as medication responders
Jasek et al. (1992)	N = 8; 8.1 (2.8) years	13-week double-blind, placebo-controlled crossover	Clonidine; 4–10 µg kg ⁻¹	Conners Abbreviated Parent-Teacher Questionnaire; ADD-H Comprehensive Teacher's Rating Scale; Aberrant Behavior Checklist—Hyperactivity subscale	Clonidine treatment associated with modest improvement of hyperactivity
Scallan et al. (2006)	N = 25; 9.03 (3.14) years	8-week open label	Guanfacine; 1.0–3.0 mg	Aberrant Behavior Checklist—Hyperactivity subscale	12/25 (48%) rated as “much improved” or “very much improved” on Clinical Global Impressions-Improvement
Handen et al. (2008)	N = 11; 5–9 years	6-week double-blind, placebo-controlled crossover	Guanfacine; 3 mg (max)	Aberrant Behavior Checklist—Hyperactivity subscale	5/11 (45%) considered medication responders
Posey et al. (2006)	N = 16; 7.7 (2.2) years	8-week open label	Atomoxetine; 1.2–1.4 mg kg ⁻¹ (target)	Clinical Global Impressions Severity and Improvement scales	12/16 (75%) rated as “much improved” or “very much improved” on Clinical Global Impressions-Improvement
Troost et al. (2006)	N = 12; 6–14 years	10-week open label	Atomoxetine; 1.19 (0.41) mg kg ⁻¹	ADHD Rating Scale	44% decrease in ADHD symptoms from baseline assessment
Arnold et al. (2006)	N = 16; 5–15 years	13-week double-blind, placebo-controlled crossover	Atomoxetine; 1.4 mg kg ⁻¹ (max)	Aberrant Behavior Checklist—Hyperactivity subscale	9/16 (56%) responded to atomoxetine; 4/16 (25%) responded to placebo
Doyle et al. (2006)	N = 8; 13.5 (2.8) years	Retrospective chart review (mean treatment length of 18 (9) weeks)	Donepezil; 2.5–30 mg	Clinical Global Impressions—Global Severity	7/8 (88%) showed improvement of ADHD symptoms on CGI Global Severity scale

ADHD = attention-deficit/hyperactivity disorder; ADD-H = attention deficit disorder with hyperactivity; MPH = methylphenidate.

tual disability. Following a single-blind methylphenidate titration, each child entered a 4-week double-blind crossover phase with 2 weeks of placebo and 2 weeks of the child's "best" methylphenidate dose (total daily doses ranging from 5.0 to 20.0 mg). Treatment with methylphenidate was reported to improve parent-rated ADHD symptoms in 50% of these children. Owing to treatment-emergent adverse events such as increased stereotypic behavior, upset stomach, sleep-related difficulties, and emotional lability, the authors recommended that this young population should be monitored closely during treatment with methylphenidate.

Amphetamine-based preparations

Despite the documented efficacy [Pliszka et al., 2007] and wide use of amphetamine-based preparations in typically-developing children with ADHD, it does not appear that amphetamine-based preparations have been specifically studied in children and adolescents with a PDD for the treatment of ADHD-like symptoms.

Clonidine and guanfacine

The use of the alpha-2 adrenergic agonists clonidine and guanfacine to treat symptoms of ADHD in children with a PDD diagnosis has been described in a few reports. These studies have generally been limited by small sample sizes and lack of a placebo control. One small double-blind, placebo-controlled crossover study of 8 male children (ages 5–11 years) with autistic disorder has evaluated the efficacy of clonidine in reducing symptoms of ADHD [Jaselskis et al., 1992]. Patients were treated for 13 weeks with clonidine or placebo at a dose range of 0.15–0.20 mg day⁻¹. A modest improvement in hyperactivity was noted with clonidine treatment, based upon parent and teacher ratings.

Posey et al. [2004] conducted a retrospective analysis of medical charts of children and adolescents (ages 3–18 years) with a PDD diagnosis to determine the response of hyperactivity, inattention, and impulsivity to treatment with guanfacine. Eighty patients were treated with a mean daily dose of 2.6 (1.7) mg guanfacine for an average of 334 (374) days. Based on ratings of "much improved" or "very much improved" on the CGI global improvement item, 23.8% of the patients were considered to be responders to treatment with guanfacine. Additionally, the authors noted symptom improvement in hyperactivity (26.7%) and inattention (21.1%).

A pilot study by Scahill et al. [2006] evaluated the efficacy of guanfacine in pediatric patients with PDD and hyperactivity. Twenty-three children (ages 5–14) received 8 weeks of open label treatment with 1.0–3.0 mg day⁻¹ guanfacine. Hyperactivity significantly decreased on parent- and teacher-rated measures. Additionally, 48% of these children were rated as "much improved" or "very much improved" on the CGI-I.

One double-blind, placebo-controlled crossover study has investigated the short-term efficacy of guanfacine (maximum dose of 3 mg day⁻¹) in children with autism and/or intellectual disabilities [Handen et al. 2008]. Eleven children (ages 5–9 years) with a diagnosis of intellectual disability and/or a PDD suffering from either comorbid ADHD or clinically significant deficits in the areas of inattention and overactivity participated in this 6-week study. Nearly half of the patients were considered to be treatment responders, based on a 50% decrease in the Aberrant Behavior Checklist Hyperactivity subscale score.

Atomoxetine

Two open-label studies of children with PDD and symptoms of ADHD have reported considerable benefits from treatment with atomoxetine [Posey et al., 2006; Troost et al., 2006]. Posey et al. [2006] provided 16 children and adolescents (ages 6–14 years) with a PDD diagnosis and a chief complaint of inattention, hyperactivity, or impulsivity with open label atomoxetine [mean dose of 1.2 (0.3) mg kg⁻¹ day⁻¹] for 8 weeks. Twelve participants were rated as "much improved" or "very much improved" on the CGI-I, and a significant improvement of ADHD symptoms on the SNAP-IV and the Aberrant Behavior Checklist were noted. Similarly, Troost et al. [2006] treated 12 pediatric patients (ages 6–14 years) with PDD accompanied with symptoms of ADHD with open label atomoxetine for 10 weeks. Treatment with atomoxetine [1.9 (0.41) mg kg⁻¹ day⁻¹] reduced ADHD symptoms, based upon scores on the ADHD Rating Scale and the Conners Parent Rating Scale-Revised. Atomoxetine was generally well tolerated in both trials.

One placebo-controlled crossover study by Arnold et al. [2006] has evaluated the efficacy of atomoxetine. Sixteen children (ages 5–15 years) diagnosed with an autism spectrum disorder and prominent ADHD symptoms were randomly assigned to treatment with

atomoxetine [44.2 (21.9) mg day⁻¹] or placebo (48.0 mg day⁻¹) for 6 weeks each. The authors noted that treatment with atomoxetine was superior to placebo on the primary outcome measure, the Hyperactivity subscale of the Aberrant Behavior Checklist. Nine of the 16 patients were considered responders to treatment with atomoxetine, according to ratings of "much improved" or "very much improved" on the CGI-I.

Possible treatment options considered in case series/case reports

In 2006, Doyle et al. published a series of cases in which eight youth (ages 10–17 years) with PDD and symptoms of ADHD were treated with donepezil, a reversible acetylcholinesterase inhibitor. Donepezil likely was chosen for its documented ability to enhance cognition in Alzheimer's disease-afflicted patients. Daily doses of donepezil ranged from 2.5 to 30 mg for a mean of 18 (9) weeks. The authors noted that most patients experienced improvement in ADHD-like symptomatology. In fact, six of the eight patients were rated as "much improved" or "very much improved" on the Clinical Global Impressions-Severity scale at the end of treatment with donepezil. Additionally, two case reports of children with autistic disorder treated with ziprasidone for ADHD symptoms have been published [Goforth and Rao, 2003; Duggal, 2007]. Goforth and Rao [2003] described the effects of a low-dose (10 mg day⁻¹ given nightly) of ziprasidone in a 7-year-old boy with autistic disorder. After 8 weeks of treatment with ziprasidone, the patient demonstrated marked improvement in behavior and attention. Another case report of a 15-year-old boy with autistic disorder, ADHD symptoms, and moderate mental retardation described improvements in attention, distractibility, hyperactivity, and impulsivity when 120 mg day⁻¹ of ziprasidone was added to his current dose of methylphenidate [Duggal, 2007].

Genetic Disorders and Prenatal Exposure to Ethanol

Similar to the treatment of ADHD and ADHD-like symptoms of children and adolescents with subaverage intelligence and pervasive developmental disorders, such treatment in pediatric patients suffering from genetic disorders and prenatal exposure to ethanol has generally been characterized by use of stimulant medications. A summary of selected studies for the treatment of ADHD and ADHD-like

symptoms in these populations is provided in Table 3.

Fragile X syndrome

Medication targeting ADHD symptoms in children and adolescents with fragile X syndrome (FXS) has been generally effective. Chart reviews have shown that stimulants are the most frequently used and the most frequently helpful medication class; in fact, some cohorts have experienced a response rate of up to 75% [Berry-Kravis and Potanos, 2004], a rate similar to what has been described in typically developing children with ADHD [Pliszka et al., 2007]. Additionally, according to chart reviews, clonidine and guanfacine have reported about 70% efficacies in the treatment of ADHD symptoms in this population [Berry-Kravis and Potanos, 2004].

There have been very few prospective treatment studies in this population, despite the prevalence of ADHD and its symptoms in this syndrome that predominantly affects male children. In 1991, Hilton et al. published a case study of a 6-year-old boy with FXS, ADHD, primary nocturnal enuresis, and initial insomnia in which all of the child's symptoms improved with imipramine treatment at 2.4 mg kg⁻¹ day⁻¹. A few double-blind, placebo-controlled trials for the treatment of ADHD in FXS have also been published. Hagerman et al. [1988] evaluated the effectiveness of methylphenidate (0.3 mg kg⁻¹) and dextroamphetamine (0.2 mg kg⁻¹) compared to placebo in a double-blind crossover trial of 15 children with FXS. Each dose of study medication was administered twice daily for 1-week each. When the children were treated with methylphenidate, improvement was noted in socialization skills and attention span. Of the 15 children, 10 were considered responders to stimulant treatment and continued treatment with methylphenidate or dextroamphetamine posttrial.

Torrioli et al. [1999, 2008] conducted two double-blind, placebo-controlled studies of L-acetylcarnitine (LAC) in children with ADHD and FXS. In the recent trial by Torrioli et al. [2008], 63 males (ages 6–13 years) with FXS and comorbid ADHD were randomized to 1 year of treatment with either LAC (500 mg, twice daily) or placebo. According to parent and teacher ratings on the Conners Global Index (CGI), treatment with LAC effectively reduced hyperactive behavior and increased attention. Patients treated with placebo experienced decreases in hyperactive behavior on both parent

Table 3. Summary of Selected Studies for the Treatment of Comorbid ADHD and ADHD-like Symptoms in Children with Genetic Disorders and Prenatal Exposure to Ethanol

Authors (year)	Sample Size and Age Range or Mean (SD)/ Age	Diagnosis	Study Design	Medication and Total Daily Dose	Primary ADHD Outcome Measures	Improvement/Effectiveness/Response
Hagerman et al. (1988)	N = 15; 3–11 years	Fragile X	3-week double-blind, placebo-controlled crossover	Methylphenidate, 0.6 mg kg ⁻¹ ; Dextroamphetamine, 0.2 mg kg ⁻¹	Conners Abbreviated Parent-Teacher Questionnaire; ADD-H Comprehensive Teacher Rating Scale	10/15 (67%) individually judged clinical responders on either medication
Torrioli et al. (2008)	N = 63; 6–13 years	Fragile X	12-month double-blind, placebo-controlled	L-acetylcarnitine, 1,000 mg	Conners Global Index-Parents; Conners Global Index-Teachers	Strongest reduction in hyperactivity found in patients treated with L-acetylcarnitine on the Conners Global Index-Parents
Capone et al. (2008)	N = 23; 7.8 (2.6) years	Trisomy 21 plus autistic spectrum disorder	10- to 18-week open label	Risperidone, 0.25–1.5 mg	Aberrant Behavior Checklist	Most significant symptom reductions noted on Hyperactivity, Stereotypy, and Lethargy subscales
Oosterheld et al. (1998)	N = 4; 5–12 years	Fetal alcohol syndrome	3-week double-blind, placebo-controlled crossover	Methylphenidate, 1.8 mg kg ⁻¹	Conners Parent Rating Scale; Conners Teacher Rating Scale	Methylphenidate associated with significant improvement on Hyperactivity Index scale on both outcome measures

ADHD = attention-deficit/hyperactivity disorder; ADD-H = attention deficit disorder with hyperactivity.

and teacher ratings on the CGI, but at a degree significantly less than patients treated with LAC ($P = 0.05$).

Trisomy 21/Down syndrome

Approximately 6–8% of children with Down Syndrome (DS) are diagnosed with comorbid ADHD [Myers and Pueschel, 1991; Dykens, 2007]. Pharmacological treatment recommendations for children with DS and ADHD are similar to those recommendations for children with ADHD who do not suffer from DS [Capone et al., 2006]. One study of 23 children (ages 3–13 years) with DS, severe intellectual disability, and comorbid autism spectrum disorders described significant reductions in the Hyperactivity, Stereotypy, and Lethargy subscales of the Aberrant Behavior Checklist after 13 weeks of open label treatment with 0.25–1.5 mg day⁻¹ risperidone [Capone et al., 2008].

Tuberous sclerosis complex

It has been estimated that between 30 and 60% of patients with tuberous sclerosis complex (TSC) meet diagnostic criteria for ADHD [D'Agati et al., 2009]. However, there are no published controlled studies of ADHD treatment in children and adolescents with TSC. Perhaps more concerning is that up to 96% of children with TSC suffer from epilepsy, and treatment of ADHD in children and adolescents suffering from TSC can be particularly challenging [Curatolo et al., 2002]. Further, treatment with an antiepileptic may putatively aggravate a patient's ADHD symptoms, and pharmacological treatments for ADHD may putatively aggravate a patient's epilepsy [D'Agati et al., 2009; Kaufmann et al., 2009]. Therefore, when considering pharmacotherapy, a cautious approach is recommended. For further information on the treatment of epilepsy in pediatric patients with developmental disorders, please refer to the article on this topic within this edition of *Developmental Disabilities Research Reviews*.

Neurofibromatosis Type 1

It has been reported that ~40% [Koth et al., 2000; Hyman et al., 2006] to ~50% [Mautner et al., 2002] of children with Neurofibromatosis Type 1 (NF-1) meet DSM-IV diagnostic criteria for ADHD, yet there are no published double-blind, placebo-controlled treatment studies in this population. One open-label trial has evaluated the effectiveness of methylphenidate in the treatment of ADHD in children with

NF-1 [Mautner et al., 2002]. Twenty children with NF-1 and ADHD [mean age 10.7 (2.2) years] and 14 children with uncomplicated ADHD [mean age 9.9 (1.9) years] were administered the Test of Variables of Attention (TOVA) (the main outcome measure) at baseline. The TOVA, a test which uses a task with two alternating graphic symbols presented at a fixed random presentation, contains four scales: omission (participant fails to press the button); commission (participant presses the button when the non-target appears); response time (mean of the correct response time, in milliseconds); and response time variability (variability of the response time, in milliseconds). Children in the NF-1 + ADHD and the ADHD-only groups received a starting dose of 5 mg methylphenidate after administration of the baseline TOVA. TOVAs were administered after 2.5 mg methylphenidate dose increases until TOVA performance reached the "normal" range (>95) or until no further improvement on the test could be obtained after a total methylphenidate dose increase of 5 mg. The final methylphenidate dose was defined as the optimized dose, and each patient was to be given his optimized dose twice daily for 1 year. Optimized doses of methylphenidate ranged from 5 to 15 mg. The mean dose for the NF-1 + ADHD group was 7.4 (2.36) mg and 9.29 (2.67) for the ADHD-only group. After 1 year of follow-up, the NF-1 + ADHD group reported significant improvements on three of the four TOVA scales (all except for "response time variability"); the ADHD-only group improved significantly on all four scales.

Williams syndrome

Despite 65% of children with Williams syndrome being reported to meet diagnostic criteria for ADHD [Leyfer et al., 2006] and the behavioral and attention difficulties associated with Williams syndrome, there are no double-blind, placebo-controlled treatment studies in this population. Two case reports of children with Williams syndrome treated with methylphenidate have been published. Bawden et al. [1997] and Power et al. [1997] reported improvements in attention, behavior, and impulsivity after treatment with methylphenidate at doses of 5 mg twice daily and 10 mg twice daily, respectively.

Prenatal Exposure to Ethanol

Attention deficits are common difficulties in children with fetal alcohol

spectrum disorders (FASD) and ADHD is perhaps the most common comorbid mental health condition in children with FASD [Doig et al., 2008; Herman et al., 2008]. In fact, children prenatally exposed to alcohol have been reported to be at a 2.5-fold increased risk for receiving an ADHD diagnosis [Mick et al., 2002]. A few trials of stimulant therapy have described improvements in ADHD symptomatology in this population [Oosterheld et al., 1998; O'Malley et al., 2000; Coe et al., 2001]. One small randomized, double-blind, cross-over study of methylphenidate treatment in four children (ages 5–12 years) meeting diagnostic criteria for ADHD and fetal alcohol syndrome (FAS) or partial FAS reported improvements in hyperactivity/impulsivity scores on the Conners' Parent and Teacher Rating Scales [Oosterheld et al., 1998]. A retrospective analysis of 30 patients (ages 6–17 years) with FAS or an alcohol-related neurodevelopmental disorder with comorbid ADHD symptomatology reported a "positive clinical response" in 22% of the 23 patients treated with methylphenidate and in 79% of the 19 patients treated with dextroamphetamine [O'Malley et al., 2000]. Additionally, a medical chart review of medication use in 22 patients (ages 3–17 years) with an FASD reported that 63% of patients treated with stimulant medications for hyperactivity, inattention, or impulsivity responded well to treatment [Coe et al., 2001]. Considering the perceived benefit of stimulant treatment in this population, it may be fair to say that further study of medication treatment for ADHD and ADHD symptomatology in children suffering from an FASD is warranted.

CONCLUSIONS

The existing data for the treatment of ADHD and ADHD-like symptoms in children and adolescents with developmental disorders are of varying degrees of methodological stringency. Nevertheless, these data suggest that the agents used to treat ADHD in typically-developing children appear to have some degree of benefit when given to pediatric patients with developmental disorders. At times, however, the risks of treatment including the side effect burden for some children within these populations appear to be greater than in typically-developing children. Therefore, it is recommended that, for the treatment of ADHD and ADHD-like symptoms in children with developmental disorders, the literature be

followed when it exists and that the limitations of methodological stringency be appreciated. In the absence of data, the agents used in typically-developing children should be considered. However, it is important to recognize the putative reduced benefits and putative increased risks in these patient populations. Therefore, when considering pharmacotherapy, a cautious approach of starting with a low medication dose and a slow upward titration is recommend, particularly owing to the vulnerability of the population and the relative paucity of data.

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REFERENCES

Agarwal V, Sitholey P, Kumar S, et al. 2001. Double-blind, placebo-controlled trial of clonidine in hyperactive children with mental retardation. *Ment Retard* 39:259–267.

Aman MG, Farmer CA, Hollway J, et al. 2008. Treatment of inattention, overactivity, and impulsiveness in autism spectrum disorders. *Child Adolesc Psychiatr Clin N Am* 17:713–738.

Aman MG, Kern RA, McGhee DE, et al. 1993. Fenfluramine and methylphenidate in children with mental retardation and ADHD: clinical and side effects. *J Am Acad Child Adolesc Psychiatry* 32:851–859.

Aman MG, Marks RE, Turbott SH, et al. 1991. Clinical effects of methylphenidate and thioridazine in intellectually subaverage children. *J Am Acad Child Adolesc Psychiatry* 30:246–256.

Arnold LE, Aman MG, Cook AM, et al. 2006. Atomoxetine for hyperactivity in autism spectrum disorders: placebo-controlled crossover pilot trial. *J Am Acad Child Adolesc Psychiatry* 45:1196–1205.

Bawden HN, MacDonald GW, Shea S. 1997. Treatment of children with Williams syndrome with methylphenidate. *J Child Neurol* 12:248–252.

Berry-Kravis E, Potanos K. 2004. Psychopharmacology in fragile X syndrome—present and future. *Ment Retard Dev Disabil Res Rev* 10:42–48.

Capone GT, Goyal P, Ares W, et al. 2006. Neurobehavioral disorders in children, adolescents, and young adults with Down syndrome. *Am J Med Genet C Semin Med Genet* 142C:158–172.

Capone GT, Goyal P, Grados M, et al. 2008. Risperidone use in children with Down syn-

drome, severe intellectual disability, and comorbid autistic spectrum disorders: a naturalistic study. *J Dev Behav Pediatr* 29:106–116.

Coe J, Sidders J, Riley K, et al. 2001. A survey of medication responses in children and adolescents with fetal alcohol syndrome. *Ment Health Aspects Dev Disabil* 4:148–155.

Conner DF, Rubin J. 2010. Guanfacine extended release in the treatment of attention deficit hyperactivity disorder in children and adolescents. *Drugs Today (Barc)* 46:299–314.

Correia Filho AG, Bodanese R, Silva TL, et al. 2005. Comparison of risperidone and methylphenidate for reducing ADHD symptoms in children and adolescents with moderate mental retardation. *J Am Acad Child Adolesc Psychiatry* 44:748–755.

Curatolo P, Verdecchia M, Bombardieri R. 2002. Tuberous sclerosis complex: a review of neurological aspects. *Eur J Paediatr Neurol* 6:15–23.

D'Agati E, Moavero R, Cerminara C, et al. 2009. Attention-deficit hyperactivity disorder (ADHD) and tuberous sclerosis complex. *J Child Neurol* 24:1282–1287.

Doig J, McLennan JD, Gibbard WB. 2008. Medication effects on symptoms of attention-deficit/hyperactivity disorder in children with fetal alcohol spectrum disorder. *J Child Adolesc Psychopharmacol* 18:365–371.

Doyle RL, Frazier J, Spencer TJ, et al. 2006. Donepezil in the treatment of ADHD-like symptoms in youths with pervasive developmental disorder: a case series. *J Atten Disord* 9:543–549.

Duggal HS. 2007. Ziprasidone for maladaptive behavior and attention-deficit/hyperactivity disorder symptoms in autistic disorder. *J Child Adolesc Psychopharmacol* 17:261–263.

Dykens EM. 2007. Psychiatric and behavioral disorders in persons with Down syndrome. *Ment Retard Dev Disabil Res Rev* 13:272–278.

Findling RL. 2008. Evolution of the treatment of attention-deficit/hyperactivity disorder in children: a review. *Clin Ther* 30:942–957.

Frazier JA, Biederman J, Bellordre CA, et al. 2001. Should the diagnosis of attention-deficit/hyperactivity disorder be considered in children with pervasive developmental disorder? *J Atten Disord* 4:203–211.

Ghuman JK, Aman MG, Lecavalier L, et al. 2009. Randomized, placebo-controlled, crossover study of methylphenidate for attention-deficit/hyperactivity disorder symptoms in pre-schoolers with developmental disorders. *J Child Adolesc Psychopharmacol* 19:329–339.

Goforth HW, Rao MS. 2003. Improvement in behavior and attention in an autistic patient treated with ziprasidone. *Aust N Z J Psychiatry* 37:775–776.

Hagerman RJ, Murphy MA, Wittenberger MD. 1988. A controlled trial of stimulant medication in children with the fragile X syndrome. *Am J Med Genet* 30:377–392.

Handen BL, Breaux AM, Gosling A, et al. 1990. Efficacy of methylphenidate among mentally retarded children with attention deficit hyperactivity disorder. *Pediatrics* 86:922–930.

Handen BL, Breaux AM, Janosky J, et al. 1992. Effects and noneffects of methylphenidate in children with mental retardation and ADHD. *J Am Acad Child Adolesc Psychiatry* 31:455–461.

Handen BL, Feldman HM, Lurier A, et al. 1999. Efficacy of methylphenidate among pre-school children with developmental disabil-

ities and ADHD. *J Am Acad Child Adolesc Psychiatry* 38:805–182.

Handen BL, Johnson CR, Lubetsky M. 2000. Efficacy of methylphenidate among children with autism and symptoms of attention-deficit hyperactivity disorder. *J Autism Dev Disord* 30:245–255.

Handen BL, McAuliffe S, Janosky J, et al. 1995. Methylphenidate in children with mental retardation and ADHD: effects on independent play and academic functioning. *J Dev Phys Disabil* 7:91–103.

Handen BL, Sahl R, Hardan AY. 2008. Guanfacine in children with autism and/or intellectual disabilities. *J Dev Behav Pediatr* 29:303–308.

Herman LE, Acosta MC, Chang PN. 2008. Gender and attention deficits in children diagnosed with a fetal alcohol spectrum disorder. *Can J Clin Pharmacol* 15:e411–e419.

Hilton DK, Martin CA, Heffron WM, et al. 1991. Imipramine treatment of ADHD in a fragile X child. *J Am Acad Child Adolesc Psychiatry* 30:831–834.

Hyman SL, Shores EA, North KN. 2006. Learning disabilities in children with neurofibromatosis type 1: subtypes, cognitive profile, and attention-deficit-hyperactivity disorder. *Dev Med Child Neurol* 48:973–977.

Jaselskis CA, Cook EH Jr, Fletcher KE, et al. 1992. Clonidine treatment of hyperactive and impulsive children with autistic disorder. *J Clin Psychopharmacol* 12:322–327.

Kaufmann R, Goldberg-Stern H, Shuper A. 2009. Attention-deficit disorders and epilepsy in childhood: incidence, causative relations and treatment possibilities. *J Child Neurol* 24:727–733.

Koth CW, Cutting LE, Denckla MB. 2000. The association of neurofibromatosis type 1 and attention deficit hyperactivity disorder. *Child Neuropsychol* 6:185–194.

Kratochvil CJ, Vaughan BS, Barker A, et al. 2009. Review of pediatric attention deficit/hyperactivity disorder for the general psychiatrist. *Psychiatr Clin N Am* 32:39–56.

Leskovec TJ, Rowles BM, Findling RL. 2008. Pharmacological treatment options for autism spectrum disorders in children and adolescents. *Harv Rev Psychiatry* 16:97–112.

Leyfer OT, Woodruff-Borden J, Klein-Tasman BP, et al. 2006. Prevalence of psychiatric disorders in 4 to 16-year-olds with Williams syndrome. *Am J Med Genet B Neuropsychiatr Genet* 141B:615–622.

Mautner VF, Kluwe L, Thakker SD, et al. 2002. Treatment of ADHD in neurofibromatosis type 1. *Dev Med Child Neurol* 44:164–170.

Mick E, Biederman J, Faraone SV, et al. 2002. Case-control study of attention-deficit/hyperactivity disorder and maternal smoking, alcohol use, and drug use during pregnancy. *J Am Acad Child Adolesc Psychiatry* 14:378–385.

Myers BA, Pueschel SM. 1991. Psychiatric disorders in persons with Down syndrome. *J Nerv Ment Dis* 179:609–613.

O'Malley KD, Koplin B, Dohner VA. 2000. Psychostimulant clinical response in fetal alcohol syndrome. *Can J Psychiatry* 45:90–91.

Oosterheld JR, Kofoed L, Tervo R, et al. 1998. Effectiveness of methylphenidate in Native American children with fetal alcohol syndrome and attention deficit/hyperactivity disorder: a controlled pilot study. *J Child Adolesc Psychopharmacol* 8:39–48.

Payton JB, Burkhart JE, Hersen M, et al. 1989. Treatment of ADDH in mentally retarded children: a preliminary study. *J Am Acad Child Adolesc Psychiatry* 28:761–767.

- Pearson DA, Santos CW, Roache JD, et al. 2003. Treatment effects of methylphenidate on behavioral adjustment in children with mental retardation and ADHD. *J Am Acad Child Adolesc Psychiatry* 42:209–216.
- Pliszka S;AACAP Work Group on Quality Issues. 2007. Practice parameter for the assessment and treatment of children and adolescents with attention-deficit/hyperactivity disorder. *J Am Acad Child Adolesc Psychiatry* 46: 894–921.
- Posey DJ, Puntney JI, Sashner TM, et al. 2004. Guanfacine treatment of hyperactivity and inattention in pervasive developmental disorders: a retrospective analysis of 80 cases. *J Child Adolesc Psychopharmacol* 14:233–241.
- Posey DJ, Wiegand RE, Wilkerson J, et al. 2006. Open-label atomoxetine for attention-deficit/hyperactivity disorder symptoms associated with high-functioning pervasive developmental disorders. *J Child Adolesc Psychopharmacol* 16:599–610.
- Power TJ, Blum NJ, Jones SM, et al. 1997. Brief report: response to methylphenidate in two children with Williams syndrome. *J Autism Dev Disord* 27:79–87.
- Quintana H, Birmaher B, Stedje D, et al. 1995. Use of methylphenidate in the treatment of children with autistic disorder. *J Autism Dev Disord* 25:283–294.
- Research Units on Pediatric Psychopharmacology (RUPP) Autism Network. 2005. Randomized, controlled, crossover trial of methylphenidate in pervasive developmental disorders with hyperactivity. *Arch Gen Psychiatry* 62:1266–1274.
- Scahill L, Aman MG, McDougle CJ, et al. 2006. A prospective open trial of guanfacine in children with pervasive developmental disorders. *J Child Adolesc Psychopharmacol* 16:589–598.
- Torrioli MG, Vernacotola S, Mariotti P, et al. 1999. Double-blind, placebo-controlled study of L-acetylcarnitine for the treatment of hyperactive behavior in fragile X syndrome. *Am J Med Genet* 87:366–368.
- Torrioli MG, Vernacotola S, Peruzzi L, et al. 2008. A double-blind, parallel, multicenter comparison of L-acetylcarnitine with placebo on the attention deficit hyperactivity disorder in fragile X syndrome boys. *Am J Med Genet A* 146:803–812.
- Troost PW, Steenhuis MP, Tuynman-Qua HG, et al. 2006. Atomoxetine for attention-deficit/hyperactivity disorder symptoms in children with pervasive developmental disorders: a pilot study. *J Child Adolesc Psychopharmacol* 16:611–619.
- Wigal T, Greenhill L, Chuang S, et al. 2006. Safety and tolerability of methylphenidate in preschool children with ADHD. *J Am Acad Child Adolesc Psychiatry* 45:1294–1303.
- Yoshida Y, Uchiyama T. 2004. The clinical necessity for assessing attention deficit/hyperactivity disorder (AD/HD) symptoms in children with high-functioning pervasive developmental disorder (PDD). *Eur Child Adolesc Psychiatry* 13:307–314.